

University: University of Karachi
Faculty: Faculty of Science
Department: Computer Science (UBIT)
Program: BS (Software Engineering)
Course Code: SE-453
Course Title: Software Requirements Engineering
Credit Hours: 3+0 (Theory: 03 | Lab: 00)
Semester / Year: I (1st Semester), Year: 2026
Academic Session: Fall Semester
Course Instructor(s): Sir Hussain (Assistant Professor)

Prepared by: Course Incharge _____

Approved by: Chairman _____

SECTION 1: COURSE BASIC INFORMATION

1. **Course Title:** Software Requirement Engineering
2. **Course Code:** SE-453
3. **Program Name:** BS (Software Engineering)
4. **Semester Offered:** I (1st Semester), Year: 2026
5. **Credit Hours:** 3+0 (Theory: 03 | Lab: 00)
6. **Pre-requisite(s):** Software Engineering
7. **Course Instructor(s):** Sir Hussain (Assistant Professor)
8. **Course Coordinator:** Nil

SECTION 2: COURSE DESCRIPTION

Course Contents:

1. Introduction to Requirements Engineering
2. Software Requirements, classification of requirements
3. Requirements process, Levels/layers of requirements
4. Requirement characteristics, Analyzing quality requirements
5. Software requirements in the context of systems engineering
6. Requirement evolution, requirement traceability, requirement prioritization
7. Trade-off analysis
8. Risk analysis and impact analysis
9. Requirement management
10. Interaction between requirement and architecture
11. Requirement elicitation
12. Elicitation sources and techniques
13. Requirement specification and documentation
14. Specification sources and techniques
15. Requirements validation and techniques
16. Management of Requirements
17. Introduction to Management
18. Requirements Management Problems
19. Managing Requirements in an Acquisition Organization
20. Supplier Organizations
21. Product Organizations
22. Requirements engineering for agile methods

Suggested Books:

- Software Requirements, Wiegers K. & Beatty J.
- Requirements Engineering, Elizabeth Hull, Ken Jackson and Jeremy Dick
- Requirements Engineering and Management for Software Development Projects, Chemuturi

SECTION 3: COURSE LEARNING OUTCOMES (CLOs)

- Learn about parallel and distributed computers.
- Write portable programs for parallel or distributed-architectures using Message-Passing Interface (MPI) Library
- Analyze complex problems with shared memory-programming with openMP

SECTION 5: COURSE CONTENT & WEEKLY TEACHING PLAN**Course Calendar (Morning)**

Week-01:	Introduction to Requirements Engineering Software Requirements, classification of requirements
Week-02:	Requirements process, Levels/layers of requirements Requirement characteristics, Analyzing quality requirements
Week-03:	Software requirements in the context of systems engineering Requirement evolution, requirement traceability, requirement
Week-04:	prioritization Trade-off analysis
Week-05:	Risk analysis and impact analysis Requirement management
Week-06:	Interaction between requirement and architecture Requirement elicitation
Week-07:	Elicitation sources and techniques Requirement specification and documentation
Week-08:	Course Review
Week-09:	Specification sources and techniques Requirements validation and techniques
Week-10:	Management of Requirements Introduction to Management
Week-11:	Requirements Management Problems Managing Requirements in an Acquisition Organization
Week-12:	Supplier Organizations Product Organizations
Week-13:	Requirements engineering for agile methods
Week-14:	Course Review

SECTION 6: TEACHING & LEARNING STRATEGIES

- Lectures
- Problem-Based Learning
- Lab Demonstrations
- Case Studies
- Assignments / Projects
- LMS / Google Classroom

SECTION 7: ASSESSMENT PLAN & WEIGHTAGE

<u>Assessment Tool</u>	<u>Weight (%)</u>
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Sessional	40 %
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- Quiz(s)
- Assignment(s)
- Project
- Lab Work

Final Exam	60 %
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Total	100%
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SECTION 15: REFERENCES & LEARNING RESOURCES

Textbooks

- Software Requirements, Wiegers K. & Beatty J.
- Requirements Engineering, Elizabeth Hull, Ken Jackson and Jeremy Dick
- Requirements Engineering and Management for Software Development Projects, Chemuturi

SECTION 16: APPROVAL & SIGNATURES

Course Instructor: Sir Hussain (Assistant Professor) _____

Head of Department: Dr. Sadiq Ali Khan (Chairman) _____